

LINUX LAB-1

Que-3

Que 3: Create a College directory hierarchy with subdirectories (CSE, ECE, ME, Civil), nested semester/subject folders, and display it using tree, pwd, cd, mkdir, rmdir, ls.

Ans 3: Linux provides commands for creating, navigating and displaying directories. The `mkdir` command creates directories, while the `-p` option allows creation of missing parent directories. The `cd` command is used to navigate between directories and `pwd` displays the current working directory. The `tree` command displays the complete directory hierarchy in a visual tree format. The `ls` command lists directory contents, while `rmdir` removes an empty directory. These commands were used to create and verify the College directory hierarchy containing CSE, ECE, ME and Civil departments with nested semester and subject folders.

Result

The College directory hierarchy was successfully created with CSE, ECE, ME and Civil departments, along with nested

semester and subject directories. The hierarchy was verified using `tree`, `pwd` and `ls`, and an empty directory was successfully removed using `rmdir`.

```
root@mshuaibsi: /home/LabF  X + v
root@mshuaibsi:/home/LabFiles# mkdir -p College/{CSE,ECE,ME,Civil}/{Sem1,Sem2}/{Subject1,Subject2}
root@mshuaibsi:/home/LabFiles# pwd
/home/LabFiles
root@mshuaibsi:/home/LabFiles# cd College/CSE/Sem1
root@mshuaibsi:/home/LabFiles/College/CSE/Sem1# pwd
/home/LabFiles/College/CSE/Sem1
root@mshuaibsi:/home/LabFiles/College/CSE/Sem1# cd -
/home/LabFiles
root@mshuaibsi:/home/LabFiles# tree College
College
├── CSE
│   ├── Sem1
│   │   ├── Subject1
│   │   └── Subject2
│   └── Sem2
│       ├── Subject1
│       └── Subject2
├── Civil
│   ├── Sem1
│   │   ├── Subject1
│   │   └── Subject2
│   └── Sem2
│       ├── Subject1
│       └── Subject2
├── ECE
│   ├── Sem1
│   │   ├── Subject1
│   │   └── Subject2
│   └── Sem2
│       ├── Subject1
│       └── Subject2
└── ME
    ├── Sem1
    │   ├── Subject1
    │   └── Subject2
    └── Sem2
        ├── Subject1
        └── Subject2

29 directories, 0 files
root@mshuaibsi:/home/LabFiles# rmdir College/Civil/Sem2/Subject2
root@mshuaibsi:/home/LabFiles# ls
College      combined.txt  error_log.txt  file2_renamed.txt  out.txt
apple_report.txt  err.txt      file1.txt      full_copy.txt      sample_data.txt
root@mshuaibsi:/home/LabFiles#
```